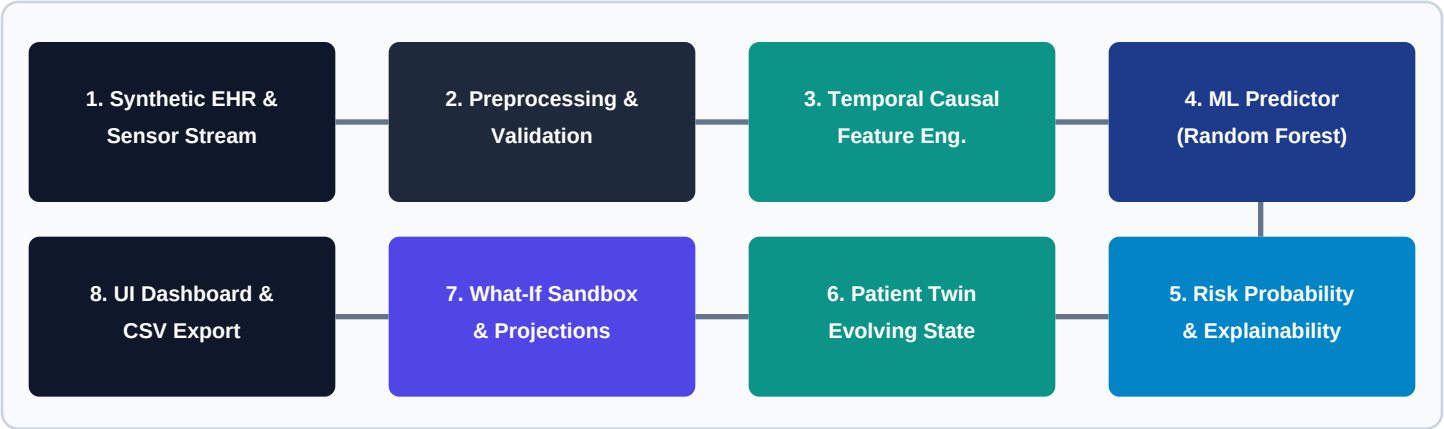
VitalTwin AI — System Architecture Document

An Explainable Digital Twin for Predictive Healthcare | Happiest Health Digital Twin Challenge 2026

Notice: Research Proof-of-Concept. Built strictly on synthetic physiological time series. Not a certified medical device.

1. High-Level Data Flow and Component Pipeline



2. Detailed Pipeline Specifications

Component	Responsibilities & Technical Formulation	Integrity & Safety Guarantees
1. Synthetic EHR & Sensor Stream	Simulates continuous 5-minute CGM interstitial glucose, PPG heart rate, step count, biphasic meal carbohydrate absorption, and sleep records across 3 patient phenotypes.	Synthetic only. Grounded in physiological gut absorption and diurnal circadian equations.
2. Data Validation & Preprocessing	Validates chronological timestamp continuity, handles missing values, sanitizes sensor bounds, and rejects corrupt sensor inputs.	Eliminates out-of-order records and guards against missing field anomalies.
3. Temporal Feature Engineering	Computes rolling glucose statistics (15m, 30m, 60m), glycemic velocity (trend slopes), rolling glucose standard deviation, accumulated carbs, steps, and EHR baselines.	**Strict Causality:** Evaluates strictly at $t' \leq t$. Zero forward data leakage.
4. ML Predictor Engine	Calibrated Random Forest classifier estimating the probability of crossing high-glucose threshold (≥ 180.0 mg/dL) within the next 120 minutes.	Trained on chronological splits with an enforced 2-hour purge gap to prevent window overlap.
5. Explainability & What-If Sandbox	Ranks top contributing features for each prediction and runs isolated hypothetical scenarios (modified carbs, exercise steps, sleep) without mutating patient ground truth.	Clear disclaimer; non-prescriptive research simulation only.

3. Held-Out Evaluation Summary

The Random Forest model was evaluated on a held-out test partition consisting of **846** consecutive 5-minute windows with positive target prevalence of **41.6%**.

• **Accuracy:** 94.1% | • **Precision:** 100.0% | • **Recall:** 85.8% | • **F1-Score:** 92.4% | • **ROC-AUC:** 0.974

4. Known Limitations and Governance

• **Synthetic Domain Gap:** In silico models cannot account for unmodeled biological phenomena (infections, systemic inflammation, pharmacokinetics variation, hormonal stress).
• **Sensor Fidelity:** Synthetic CGM curves lack sensor compression artifacts and calibration drift.
• **Regulatory Status:** For investigative proof-of-concept demonstration only.